Quadratic Inequalities

Exam code: 9MA0

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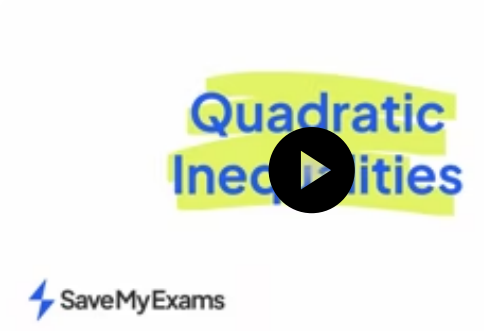


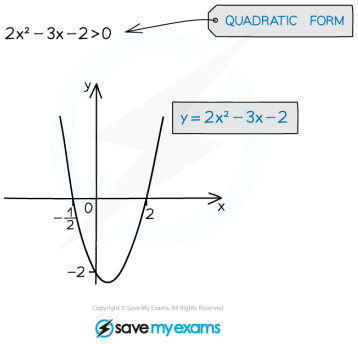
Did this video help you? Yes No

Help me understand

Quadratic inequalities

What are quadratic inequalities?

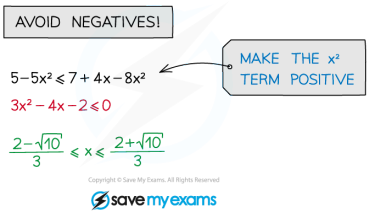
- Similar to quadratic equations, quadratic inequalities just mean there is a range of values that satisfy the solution
- Sketching a quadratic graph is essential
- Can involve the **discriminant** or **applications** in **mechanics** and **statistics**



Stuck on something?
Ask questions and get help with exactly what you're revising.
Got it

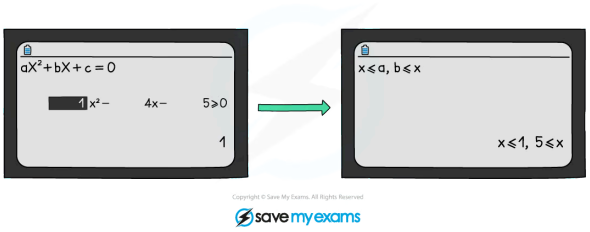
How do I solve quadratic inequalities?

- **STEP 2**: Find the **roots** of the quadratic equation
  - **Solve** $ax^2 + bx + c = 0$ to get $\boldsymbol{x_1}$ and $\boldsymbol{x_2}$ where $\boldsymbol{x_1 < x_2}$
- **STEP 3**: **Sketch** a graph of the quadratic and **label the roots**
  - As the squared term is positive it will be **"U" shaped**
- **STEP 4**: **Identify** the **region** that satisfies the inequality
  - For $\boldsymbol{ax^2 + bx + c > 0}$ you want the region **above** the *x*-axis
    - The solution is $\boldsymbol{x < x_1}$ **or** $\boldsymbol{x > x_2}$
  - For $\boldsymbol{ax^2 + bx + c < 0}$ you want the region below the *x*-axis
    - The solution is $\boldsymbol{x > x_1}$ **and** $\boldsymbol{x < x_2}$
    - This is more commonly written as $\boldsymbol{x_1 < x < x_2}$
- **avoid** multiplying or dividing by a negative number
  if unavoidable, "**flip**" the inequality sign so $\boldsymbol{< \rightarrow >}$, $\boldsymbol{\geq \rightarrow \leq}$, etc
  - **avoid** multiplying or dividing by a **variable** (**x**) that **could** be negative
    (multiplying or dividing by $\boldsymbol{x^2}$ guarantees positivity (unless **x** could be 0) but this can create extra, invalid solutions)
  - **do** rearrange to make the x^2 term positiveBe careful:



How do I solve quadratic inequalities on a calculator?

- Be aware of unconventional ways calculators can display an answer
  eg **8 > x > 2** rather than **2 < x < 8**



Examiner Tips and Tricks

- is it a “show that” or “prove” question?
  - how many marks?
  - how long is the question?

Worked Example

Find the set of values for which $3x^2 + 2x - 6 > x^2 + 4x - 2$ giving your answer in set notation.

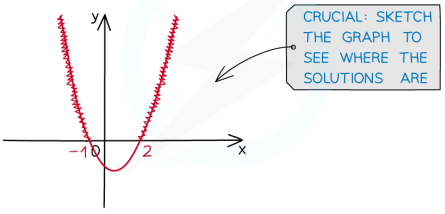
$3x^2 + 2x - 6 - x^2 - 4x + 2 > 0$

$2x^2 - 2x - 4 > 0$ ← REARRANGE TO QUADRATIC FORM

$2(x^2 - x - 2) > 0$

$x^2 - x - 2 > 0$

$(x-2)(x+1) > 0$



$x<-1$ OR $x>2$

$\{x : x<-1\} \cup \{x : x>2\}$

FINAL ANSWER IN SET NOTATION

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